

# Alexandru Buburuzan

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[alexbubu.com](http://alexbubu.com) · [google scholar](https://scholar.google.com/citations?user=alexbubu) · [github.com/alexbuburuzan](https://github.com/alexbuburuzan) · [linkedin.com/in/alexbubu](https://www.linkedin.com/in/alexbubu)

## Education

### University of Oxford · Visual Geometry Group (VGG)

*DPhil (PhD) Autonomous Intelligent Machines and Systems*

Oxford, UK · Oct 2025 – present

- Supervisor: [Prof. Andrew Zisserman](#); co-supervisor: [Dr. Puneet Dokania](#).
- Research on generative modelling for perception; fully funded by University of Oxford's [AIMS CDT](#) and [Five AI](#).

### University of Manchester

*BSc (Hons) Artificial Intelligence with Industrial Experience*

Manchester, UK · Sep 2021 – Jun 2025

- Dissertation supervised by [Prof. Tim Cootes](#): “[Reference-Guided Diffusion Inpainting for Multimodal Counterfactual Generation](#).”
- Ranked 1<sup>st</sup>/270+ in year 1, **Top 1%** in year 2; Golden Anniversary and Netcraft awards (see [full transcript](#)).
- Conferences: [CVPR'25](#), [MICCAI'23](#). Summer schools: [Oxford ML](#) (2023), [Cambridge AI Med](#) (2022), [EEML](#) (2022).
- Selected courses: Machine Learning, AI & Games, Knowledge-Based AI, Multivariate Statistics, Visual Computing.

## Employment

### Five AI – acquired by Bosch

*Research Scientist Intern*

Cambridge & Oxford, UK · May 2025 – Mar 2026

- Co-authored a survey [1], published in TMLR, of 37 foundation models for trajectory planning in autonomous driving.
- Paper under review on efficient Vision Language Action (VLA) models.

*Research Engineer Intern*

Cambridge, UK · Jun 2023 – Jun 2024

- [Main project](#): developed a new diffusion model for camera-lidar object inpainting in driving scenes [2] – [patent](#) filed.
- Fine-tuned Paint-by-Example for multimodal generation with 3D control, resulting in **13% LPIPS improvement**.
- *Other project*: co-authored a paper [3] on multimodal sensor fusion for 3D object detection in autonomous driving.
- Implemented explainability and interpretability techniques, demonstrating enhanced camera-lidar complementarity.

### Rayscape

*Research Engineer*

Remote · Jul 2021 – Jun 2023

- *Main projects*: developed a CE-marked algorithm for lung nodule segmentation, **deployed in over 100 hospitals**.
- Devised a method reducing the out-of-domain gap in multi-label chest X-ray classification by 28% for two shifts [4].
- Developed a nodule malignancy classifier, improving F1 score by 3% using Vision Transformers.
- *Other projects*: contributed to the statistical analysis for a clinical study published in Nature Scientific Reports [5].

*Machine Learning Intern*

Timisoara, Romania · Mar 2020 – Sep 2020

- *Main project*: developed an algorithm for detecting intracranial haemorrhages which **sped up the triaging process**.

## Experience

### The University of Manchester AI Society

*President*

Sep 2024 – Feb 2025

- Led a team of ten to organise ML workshops and introduced a new spotlight series of academic talks.
- Partnered with [Entrepreneurs First](#) to host their first UoM [event](#) and was selected as a [GirlsWhoML](#) campus coordinator.

*Workshops Executive*

Sep 2022 – Jun 2024

- Taught workshops on [computer vision](#) for medical image analysis and [self-supervised learning](#) with SimCLR.

## Selected Publications

\* indicates equal contribution

- [1] Oksuz K, **Buburuzan A**, Knittel A, Yao Y, Dokania P, “[Foundation Models for Trajectory Planning in Autonomous Driving: A Review of Progress and Open Challenges](#)”, *Transactions on Machine Learning Research (TMLR)*, Survey Certification, 2026.
- [2] **Buburuzan A**, Sharma A, Redford J, Dokania P, Mueller R, “[MObl: Multimodal Object Inpainting Using Diffusion Models](#)”, *CVPR Workshop on Data-Driven Autonomous Driving Simulation (DDADS)*, 2025.
- [3] Gunn J, Lenyk Z, Sharma A, Donati A, **Buburuzan A**, Redford J, Mueller R, “[Lift-Attend-Splat: Bird’s-eye-view camera-lidar fusion using transformers](#)”, *CVPR Workshop on Autonomous Driving (WAD)*, 2024.
- [4] Bercean B\*, **Buburuzan A\***, Birhala A, Avramescu C, Tenescu A, Marcu M, “[Breaking Down Covariate Shift on Pneumothorax Chest X-Ray Classification](#)”, *MICCAI UNSURE Workshop*, 2023.
- [5] Bercean B, Birhala A, Ardelean P, Barbulescu I, Benta M, Rasadean C, Costachescu D, Avramescu C, Tenescu A, Iarca S, **Buburuzan A**, Marcu M, Birsasteanu F, “[Evidence of a cognitive bias in the quantification of COVID-19 with CT: an artificial intelligence randomised clinical trial](#)”, *Scientific Reports*, 2023.